

# CS1400 Programming Lab – Common Programming Requirements

## Case Problem Labs

In this lab, you will continue working with your assigned Case Problem business to store, process, and search collections of related data using lists. Although each case problem has a different business scenario, all students will complete the same Common Programming Requirements.

The assignment is divided into Program Planning, Program Implementation, and Program Testing to reinforce the importance of planning, implementing, and testing a program before it is considered complete.

### Part 1 – Program Planning

Good programmers spend time planning before they begin writing code. Before you start programming, think about how your solution will be organized, what information your program will store, and which functions you will create. A few minutes spent planning often saves hours of debugging later.

Before writing any code, carefully review the program requirements and answer the following questions:

- What information will your program collect?
- Which information should be stored in one or more lists?
- Will your solution require one list or multiple lists?
- What calculations will your program perform?
- What value will the user be able to search for?
- Which functions will you create to organize your program?

Your responses do not need to be lengthy. The purpose of this activity is to help you organize your thoughts and develop a plan before you begin writing your program.

### Part 2 – Program Implementation

Using your program plan as a guide, write your program according to the following requirements.

#### Programming Requirements

##### Data Collection

- Prompt the user to enter multiple records related to your selected case problem.
- Store the collected information in one or more lists.
- Allow the user to enter an unspecified number of records.

##### List Processing

- Use loops to process the data stored in the list(s).
- Display all records entered by the user.
- Calculate a total.
- Calculate an average (when appropriate).
- Determine the largest value.
- Determine the smallest value.

### **Searching**

- Provide a way for the user to search for a specific item stored in one of the lists.
- Display an appropriate message indicating whether the item was found.

### **Program Design**

- Organize the program into logical functions.
- Use meaningful variable and function names.
- Include comments describing the purpose of each function.
- Validate user input where appropriate.
- Produce clear, well-formatted output.

## **Part 3 – Program Testing**

After completing your program, verify that it works correctly by testing all required functionality.

Submit at least five test cases that demonstrate:

- Normal program operation.
- Boundary or edge cases, where appropriate.
- Invalid input handling, if applicable.
- Correct calculations and program output.

## **Submission Requirements**

- Program Design (brief written responses to the design questions)
- Source code
- Test cases
- Code walk-through video (3–4 minutes) demonstrating program runs and explaining how string processes were used to verify the data.